COMPUTER ENGINEERING

QUEZON CITY

COURSE CODE/TITLE:	CPE 212 - Automating Server Management	**SECTION:**	CPE31S3
DATE PERFORMED:	October 29, 2026	**DATE SUBMITTED:**	October 29, 2026
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ACTIVITY NO. **Final Hands-on Skills Exam**

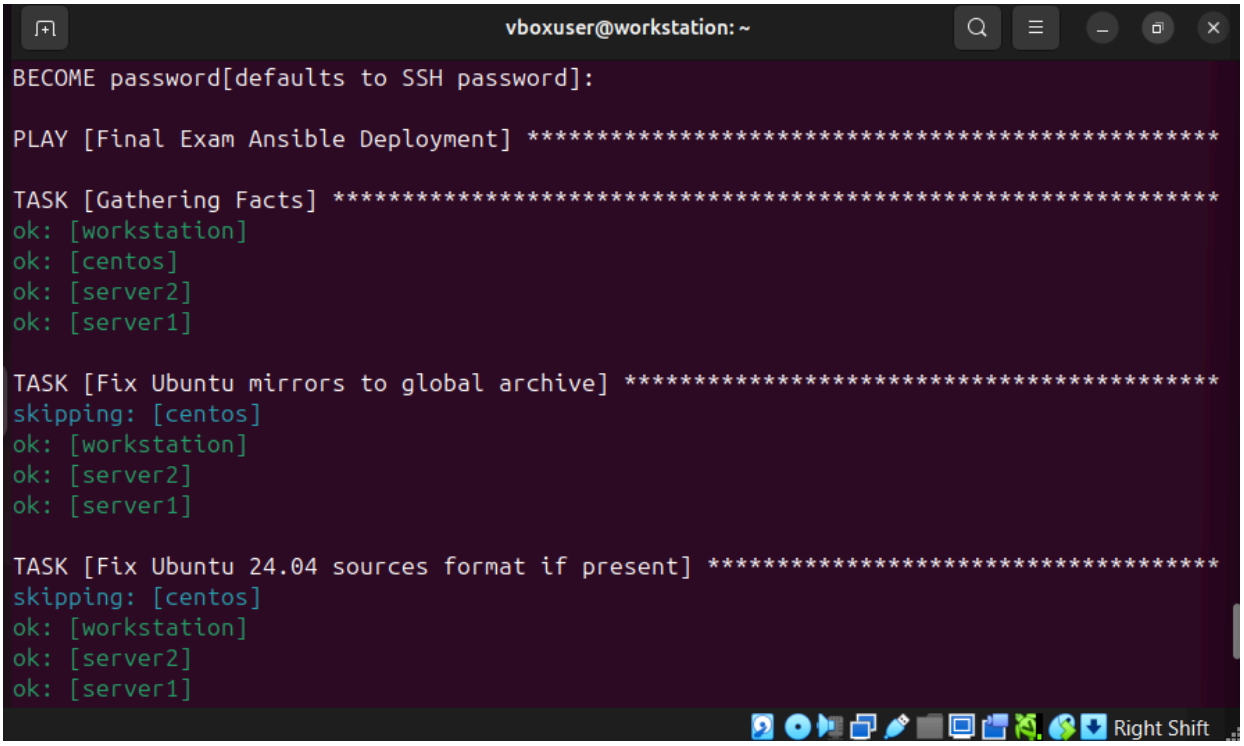
1. PROCEDURE OUTPUT

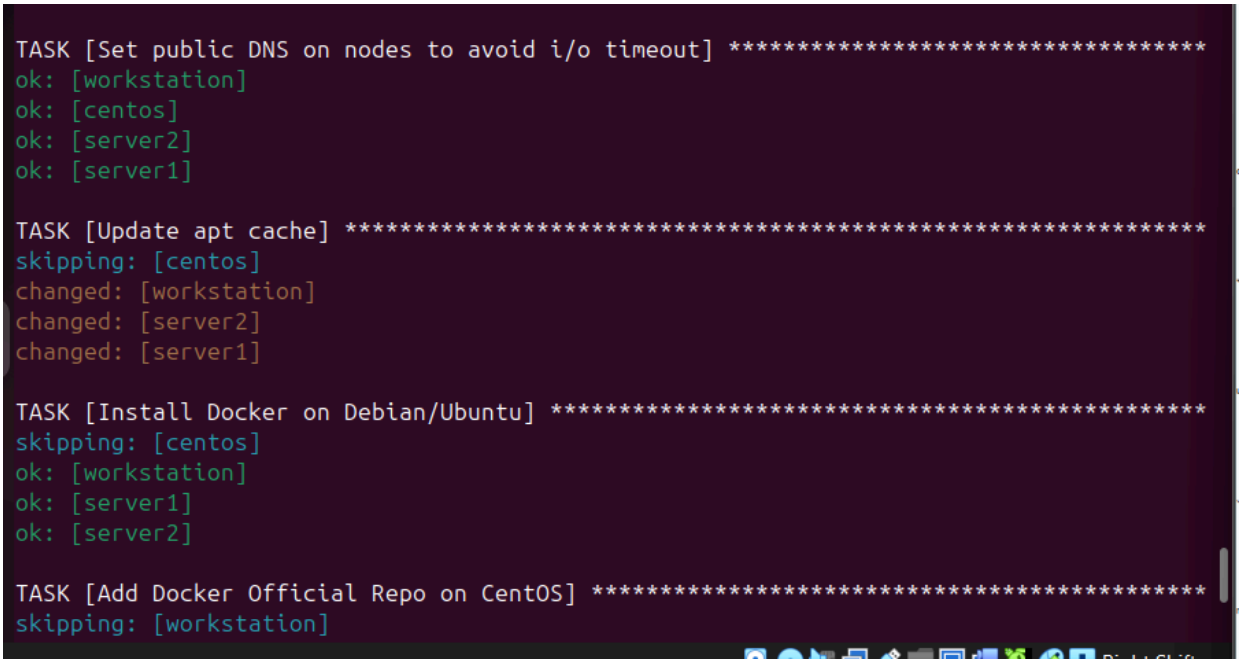
What procedures did you perform?

3. Create an Ansible playbook that does the following with an input of a config.yaml file and structure inventory file.



playbook.yml





```
TASK [Add Docker Official Repo on CentOS] ********************************************
skipping: [workstation]
skipping: [server1]
skipping: [server2]
ok: [centos]

TASK [Install Docker CE on CentOS] ***************************************************
skipping: [workstation]
skipping: [server1]
skipping: [server2]
ok: [centos]

TASK [Upgrade docker python SDK on CentOS] *******************************************
skipping: [workstation]
skipping: [server1]
skipping: [server2]
ok: [centos]

TASK [Ensure Docker service is running] **********************************************
ok: [workstation]
```

```
TASK [Ensure Docker service is running] **********************************************
ok: [workstation]
ok: [centos]
ok: [server2]
ok: [server1]

TASK [Deploy Enterprise Web Server] **************************************************
ok: [workstation]
fatal: [centos]: FAILED! => {"changed": false, "msg": "Error connecting: Error while fe
tching server API version: Not supported URL scheme http+docker"}
...ignoring
ok: [server2]
ok: [server1]

TASK [Deploy Node Exporter Container] ************************************************
ok: [workstation]
fatal: [centos]: FAILED! => {"changed": false, "msg": "Error connecting: Error while fe
tching server API version: Not supported URL scheme http+docker"}
...ignoring
```

```
tching server API version: Not supported URL scheme http+docker"}
...ignoring
ok: [server2]
ok: [server1]

TASK [Deploy Prometheus Server] ***********************************************
skipping: [server1]
skipping: [server2]
skipping: [centos]
ok: [workstation]

TASK [Set custom MOTD banner] *************************************************
ok: [workstation]
ok: [server2]
changed: [centos]
ok: [server1]

PLAY RECAP ********************************************************************
centos                     : ok=9    changed=1    unreachable=0    failed=0    skipped=
5    rescued=0    ignored=2
server1                    : ok=10   changed=1    unreachable=0    failed=0    skipped=
```

```
skipping: [server1]
skipping: [server2]
skipping: [centos]
ok: [workstation]

TASK [Set custom MOTD banner] *************************************************
ok: [workstation]
ok: [server2]
changed: [centos]
ok: [server1]

PLAY RECAP ********************************************************************
centos                     : ok=9    changed=1    unreachable=0    failed=0    skipped=
5    rescued=0    ignored=2
server1                    : ok=10   changed=1    unreachable=0    failed=0    skipped=
4    rescued=0    ignored=0
server2                    : ok=10   changed=1    unreachable=0    failed=0    skipped=
4    rescued=0    ignored=0
workstation                : ok=11   changed=1    unreachable=0    failed=0    skipped=
3    rescued=0    ignored=0
```

```
vboxuser@workstation:~$ curl http://localhost:8080
<!DOCTYPE html>
<html>
<head>
<title>Welcome to nginx!</title>
<style>
html { color-scheme: light dark; }
body { width: 35em; margin: 0 auto;
font-family: Tahoma, Verdana, Arial, sans-serif; }
</style>
</head>
<body>
<h1>Welcome to nginx!</h1>
<p>If you see this page, nginx is successfully installed and working.
Further configuration is required for the web server, reverse proxy,
API gateway, load balancer, content cache, or other features.</p>

<p>For online documentation and support please refer to
<a href="https://nginx.org/">nginx.org</a>.<br/>
To engage with the community please visit
```

```
font-family: Tahoma, Verdana, Arial, sans-serif; }
</style>
</head>
<body>
<h1>Welcome to nginx!</h1>
<p>If you see this page, nginx is successfully installed and working.
Further configuration is required for the web server, reverse proxy,
API gateway, load balancer, content cache, or other features.</p>

<p>For online documentation and support please refer to
<a href="https://nginx.org/">nginx.org</a>.<br/>
To engage with the community please visit
<a href="https://community.nginx.org/">community.nginx.org</a>.<br/>
For enterprise grade support, professional services, additional
security features and capabilities please refer to
<a href="https://f5.com/nginx">f5.com/nginx</a>.</p>

<p><em>Thank you for using nginx.</em></p>
</body>
```

```
vboxuser@workstation:~$ curl http://localhost:9100/metrics
# HELP go_gc_duration_seconds A summary of the wall-time pause (stop-the-world) duratio
n in garbage collection cycles.
# TYPE go_gc_duration_seconds summary
go_gc_duration_seconds{quantile="0"} 0
go_gc_duration_seconds{quantile="0.25"} 0
go_gc_duration_seconds{quantile="0.5"} 0
go_gc_duration_seconds{quantile="0.75"} 0
go_gc_duration_seconds{quantile="1"} 0
go_gc_duration_seconds_sum 0
go_gc_duration_seconds_count 0
# HELP go_gc_gogc_percent Heap size target percentage configured by the user, otherwise
 100. This value is set by the GOGC environment variable, and the runtime/debug.SetGCPe
rcent function. Sourced from /gc/gogc:percent.
# TYPE go_gc_gogc_percent gauge
go_gc_gogc_percent 100
# HELP go_gc_gomemlimit_bytes Go runtime memory limit configured by the user, otherwise
 math.MaxInt64. This value is set by the GOMEMLIMIT environment variable, and the runti
me/debug.SetMemoryLimit function. Sourced from /gc/gomemlimit:bytes.
# TYPE go_gc_gomemlimit_bytes gauge
```

```
ytes.
# TYPE process_virtual_memory_max_bytes gauge
process_virtual_memory_max_bytes 1.8446744073709552e+19
# HELP promhttp_metric_handler_errors_total Total number of internal errors encountered
 by the promhttp metric handler.
# TYPE promhttp_metric_handler_errors_total counter
promhttp_metric_handler_errors_total{cause="encoding"} 0
promhttp_metric_handler_errors_total{cause="gathering"} 0
# HELP promhttp_metric_handler_requests_in_flight Current number of scrapes being serve
d.
# TYPE promhttp_metric_handler_requests_in_flight gauge
promhttp_metric_handler_requests_in_flight 1
# HELP promhttp_metric_handler_requests_total Total number of scrapes by HTTP status co
de.
# TYPE promhttp_metric_handler_requests_total counter
promhttp_metric_handler_requests_total{code="200"} 0
promhttp_metric_handler_requests_total{code="500"} 0
promhttp_metric_handler_requests_total{code="503"} 0
```

Banner

```
vboxuser@workstation:~$ cat /etc/motd
Ansible Managed by Malabanan
```

2. ANSWERS TO SUPPLEMENTAL ACTIVITY

Copy and paste the questions from the lab and include your answers after.

3. ARTIFICIAL INTELLIGENCE (AI) USE DISCLOSURE STATEMENT

I hereby declare that artificial intelligence (AI) tools were used, if applicable, in the preparation of this academic work. The following indicates the nature and extent of AI assistance:

Tools Used:

Examples: ChatGPT, Microsoft Copilot, Grammarly, QuillBot, etc.

Purpose of Use:

Examples: grammar correction, idea generation, formatting, summarization, code assistance, explanation of concepts, etc.

Extent of Use:

Explain how AI was used and confirm that it did not replace original thinking, analysis, authorship, or completion of the required work.

By submitting this activity, I affirm that my use of AI complies with T.I.P.'s AI usage policy and does not exceed the permitted limits for similarity, automated content generation, or academic assistance.

I understand that any false, incomplete, or misleading disclosure may be subject to investigation in accordance with due process and may result in sanctions for violation of existing T.I.P. policies.

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